# Dave Rich

## Software Engineer

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Software Engineer with seven years of experience specializing in Go, Kubernetes, containers, and CI/CD. Over fifteen years of experience using and maintaining Linux systems. Strong engineering fundamentals with AI-powered tools enable me to build, iterate, and ship software faster than ever.

### SKILLS

**Programming**: Go (Golang), TypeScript, JavaScript, Node.js, React, PHP, Python, SQL
**Orchestration & CI/CD**: Kubernetes, Docker (containers), Jenkins, GitHub Actions, Argo CD, DevOps, GitOps
**Infrastructure & Provisioning**: Red Hat, Ubuntu, Linux, Virtualization (VMware), Ansible, Terraform, IaC
**Project Management**: Scrum, Agile, Kanban, Jira, GitHub

### WORK EXPERIENCE

**Professional Development** November 2025 – Current

Pittsburgh, PA

- Achieved Certified Kubernetes Application Developer (CKAD) certification
- Configured lab virtual environments with Proxmox to study for Certified Kubernetes Administrator (CKA) and CKAD exams
- Streamlined software development workflows using locally hosted open-source LLMs and Claude Code
- Experimented with Rust and AI to build a desktop application which optimizes rapid context switching

**Cloud Software Engineer** November 2023 – October 2025

Webbula, Wexford, PA

- Added CI to 5 core company projects including Go unit tests, static analysis, and reproducible Docker builds with GitHub actions
- Created DNS parsing library for validating SPF, DKIM, and DMARC TXT records according to RFC specifications with Go
- Used Scheduler Agent Supervisor pattern, Go, containers, and AWS SQS to create Proof of Concept monitoring system for scheduled reporting of customer DNS configuration issues
- Developed novel email address verification Proof of Concept with Python scaling up to 1 million addresses per hour
- Implemented Terraform for EmailHygiene product which reduced development and testing environment provisioning from 2 days to less than 30 minutes
- Collaborated with designer and PM to implement features for platform rewrite with Go and React
- Standardized development environments which enabling rapid onboarding of new team members for organization expansion

**Medical Leave** May 2022 – October 2023

Pittsburgh, PA

- Returned to full capacity October 2023

**Software Engineer** September 2017 – April 2022

Expedient, Pittsburgh, PA

- Developed internal ticketing & infrastructure change management application using PHP, JavaScript, SQL, and Symfony framework
- Led architecture and implementation of cloud native REST API backend microservices using Go and Gin framework
- Spearheaded adoption of Go programming language for microservices and ETL workflows which increased team velocity by 20% after 3 months
- Adopted Infrastructure as Code (IaC) practices by deploying Kubernetes clusters on Ubuntu Linux with Terraform and Rancher

- Automated deployments of Kubernetes manifests & CRDs using the GitOps pattern via ArgoCD
- Optimized multi-region ETL by leveraging Go goroutines and channels which reduced runtime by 90%
- Pioneered developer workflow using Kubernetes, Skaffold, and shell scripts which led to a speed up in provisioning developer environments from 5 days to 3 hours
- Designed architecture for multi-region high availability automated failover which enabled team to exceed uptime goal of 99.99%
- Championed adoption of Kubernetes for container orchestration resulting in a reduction of virtual infrastructure by 50%
- Conducted technical interviews with candidates for DevOps and Software Engineer roles
- Participated in 24/7 on-call rotation

**Senior Technician** April 2017 – September 2017

Expedient, Pittsburgh, PA

- Developed custom infrastructure alert processing application with Angular, TypeScript, and Node.js which reduced time to onboard new monitoring analysts by 80%
- Managed customer and internal RHEL, CentOS, and Ubuntu Linux servers by patching, troubleshooting issues, and performing online disk expansions
- Used Linux CLI (Bash) and common utilities such as awk, sed, grep, lsof, and netstat to troubleshoot issues and perform data restoration from EMC NetWorker backups
- Promoted to Senior Technician and received Q3 2017 Internal Support award from Expedient for organizational impact of custom infrastructure alert processing application

**Technician** March 2015 – April 2017

Expedient, Pittsburgh, PA

- Solved issues for managed backup environment of Windows and Linux file system, Microsoft Exchange, Microsoft SQL Server, and MySQL data for over 6,000 servers
- Responded to customer emergency by optimizing restoration of over 1600 production MSSQL databases with PowerShell to reduce downtime
- Enabled rapid onboarding of a new customer by converting over 11,000 DNS records to CSV with Node.js because a DNS zone transfer was not possible
- Received April 2016 Client Care Extra Effort Award from Expedient for quickly and efficiently resolving urgent customer issues by scripting solutions

**Analyst** August 2014 – March 2015

Expedient, Pittsburgh, PA

- Received customer support phone calls, handled ticket intake, and triaged high severity issues
- Updated virtual machine resource configurations with VMware vSphere according to customer requirements
- Earned promotion 5 months ahead of schedule for technical expertise across UNIX, Windows, backup, networking, and virtualization disciplines

## CERTIFICATIONS

**Certified Kubernetes Administrator (CKA)**

Cloud Native Computing Foundation

Expected completion November 2026

**Certified Kubernetes Application Developer (CKAD)**

Cloud Native Computing Foundation

October 2026

**AWS Certified Cloud Practitioner**

Amazon Web Services

September 2023